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Generate Topic of document: [1]In this paper author used fb-lda method and two dataset.one of them is SIGIR dataset and other 10 news group dataset. And used perplexity to find numbers. They varied the number of topics from 92 to 100 and found out 5 and 2 are ideal for this dataset. [2] In this LDA method, The cora dataset contains 2410 publications. Dataset has binary word vector 0 and 1. 0 indicates absence and 1 indicates presence. randomly have to select 9 documents from dataset and divide the vocabulary in documents into 10 topics. These topics are ideal to the cora datasets. **LDA model strategy**:[3] In gaussian HLDA method, author used 6000 abstracts from the wikipedia dataset. The initial structure of HDLA is [1,1,4,4]. Number of topics estimated in HDLA is 54 and 83.[4] In Gaussian LDA method, author has used 50 news dataset. Topic distribution of GLDA is (1,100, 1932) and full size of corpus is 1932. Found out 5 topics were ideal to the news dataset. GLDA are ranked based on density assigned by topic distribution.[5] In labelled LDA method paper, Author has used 20 corpora datasets, and reuters. They varied number of topics from 10 to 80. They found 20, 40, topics are relevent to the datasets. They trained the model and tuned it. [6] In this paper, Author has used 15 datasets. LDA is non linear method. These hyperparameters has alpha and beta. These method used perplexity with cross validation to find the number. They varied the number of topics frpm 100, 200,300 and found out 5, 10, 15 are ideal to the datasets.

References: [1] Alattar, F.; Shaalan, K. Emerging Research Topic Detection Using Filtered-LDA. AI 2021, 2, 578–599.

[2] Mahesh Korlapati, Tejaswi Ravipati, abhilash Kumar Jha. Categorizing research papers by topics using Latent dirichlet allocation model. International journal of scientific and technology research volume 8, issue 12, december 2019.

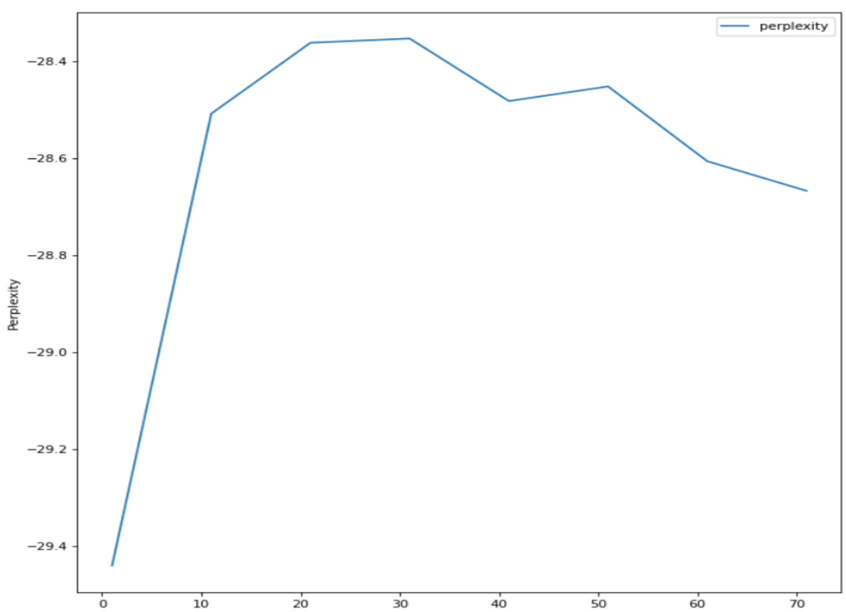
[3] Takahiro Yoshida, Ryohei Hisano, Takaaki Ohnishi , Gaussian Hierarchical Latent Dirichlet Allocation: Bringing Polysemy Back, Graduate School of Information Science and Technology The University of Tokyo February 26, 2020.

[4] Rajarshi Das, Manzil Zaheer, Chris Dyer, Gaussian LDA for Topic Models with Word Embeddings, School of Computer Science Carnegie Mellon University Pittsburgh, PA, 15213, USA. pages 3-10.

[5] Daniel Ramage, David Hall, Ramesh Nallapati and Christopher D. Manning, Labeled LDA: A supervised topic model for credit attribution in multi-labeled corpora, Computer Science Department Stanford University.

[6] Asana Neishabouri, Michel C. Desmarais, Reliability of Perplexity to Find Number of Latent Topics, The Thirty-Third International FLAIRS Conference (FLAIRS-33)

1. **<u>Datataset</u>:** Physics vs chemistry vs Biology data set has been taken from kaggle. Which has both train and testing data [1].
2. **<u>Data Pre-processing</u>:** The dataset has been preprocessed by tokenizing, removal of the stopwords and lemmatization by keeping the noun,verb and adjectives.The data was a labelled data and we took the part with maximum count of text and removed the labels and ids.
3. **<u>Problem type:</u>** Clustering.
4. **<u>Coding Language and Model Codes</u>:** The code has been done with Python and the gensim lda has been used a latent Dirichlet model. The nltk library has been used to preprocess and the gensim is used to build the final LDA model and also to preprocess the data (forming tokens).
5. **<u>Hyperparameters</u>:**The hyperparameters used in the lda model are random_state=100,update_every=1,chunksize=100,passes=10,alpha=0.3,eta= 0.03.Trial and error method has been used to select this hyperparameters.
6. **<u>Described How I arrived at optimal Number of Topic</u>:** Here the dataset has been used to convert into corpus and dictionary after being preprocessed by the above mentioned steps as the gensim lda accepts in this form. Once, the model is created the perplexity of each topic has been calculated over certain range. The topic with maximum perplexity has been considered as our final optimal topic.



Topics	The log perplexity
1	29.43
11	28.50
31	28.35
41	28.48

From the above graph we can see that the optimal topic number is 31 according to the log perplexity score.

Finally, with test set with same hyper-parameters and the optimal topic number the

Perplexity score is -19.721724995045523 (on test data).

6. **Reference:** [1] [Physics vs Chemistry vs Biology | Kaggle]